

EFM3 V3.1-R1 Correctness-Repair Research Report

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■■ INVALIDATED_BY_V3.1_R1

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The prior PR #20 headline metrics — ****DD plain SMAPE 64.84****, ****legal-oracle plain 41.39 / floor50 145.71****, and the ****A–F candidate ranking**** — are ****INVALIDATED****. They were produced by a replay that contained 8 defects (target-day DA leakage, business-day mapping error, metric-formula errors, Track D/F/C leakage & index errors, evaluation support inconsistency, and missing contract tests). All 8 are fixed in V3.1-R1. The figures below are from the ****corrected**** full 2022–2026 rolling-origin replay.

1. Scope & Decision (Forecast Availability Contract)

This patch predicts ****Real-Time (RT) prices****. The production RT circuit issues the RT forecast for day D ***before*** day D's DA clearing price is published (confirmed by the leak-free production number $RT \approx 27.4\%$). Therefore ****the target-day DA clearing price is NOT visible at RT prediction time****.

Consequence: `legal\_oos\_da\_prediction` in V3.1 was a literal copy of `da\_actual` (target-day leakage, defect #1) and is ****removed****. The only legal DA proxy is `da\_oos\_pred`, output of a genuine ****rolling-origin day-ahead model**** trained on PAST `da\_actual` only. `da\_actual` is an ACTUAL, never renamed into a prediction.

2. The 8 Defects Fixed (V3.1 → V3.1-R1)

#	Defect	V3.1-R1 fix
1	`legal_oos_da_prediction` copied `da_actual`	removed; legal proxy = rolling-origin OOS DA model (`da_oos_pred`)
2	`business_day = times.date` mapping error	reuse `utils.business_day` (D+1 00:00 → business_day D, hour 24)
3	plain SMAPE / floor50 formula errors	single canonical `fusion.metrics.plain_smape` / `smape_floor50`
4	Track D `pd.qcut(full_history)` leakage	bins fit per rolling TRAIN window only
5	Track F residual-as-input + train/infer mismatch	strict K-fold OOF two-stage (base → residual → final = base+resid)
6	Track C local/global index errors	relative→absolute index fix; assembled `C_seasonal_full`

7	B/C/DD evaluation-support inconsistency	every candidate reports coverage_rows / coverage_ratio / common mask
8	no contract tests	8 `tests/research/test_v31_*.py` (35 checks) + `run_mini_replay.py` (14 checks)

3. Methodology

- **Rolling-origin replay**: retrain every 90 days; predict each target day using only data strictly before the prediction time (STRICT_REPLAY_OOS).
- **Legal DA proxy**: `da\_oos\_pred` from a rolling-origin L2 LightGBM DA model (CPU-only; GPU disabled on this host).
- **Unified metrics**: `plain\_smape = 100·|y-█| / ((|y|+|█|)/2)`; `smape\_floor50` with denominator floored at 50 (tail-weighted).
- **Unified evaluation support**: each candidate reports coverage; the final ranking uses ONE full-coverage common mask. Intrinsically-partial candidates (per-season C_*, B_midday 9-16-only) are excluded from that mask by design.
- **Legal Oracle**: EX_POST_ACTUAL_AWARE_UPPER_BOUND — per-row min-loss selection among candidates; invariants assert selected==one candidate, oracle loss ≤ each candidate loss, row count == common mask.

4. Full Replay — Calibration Baselines

Candidate	plain SMAPE	floor50 SMAPE	cov.ratio	Δ vs DD (plain)	9-16 plain
A05_huber	57.02	56.43	0.94	-13.15	95.91
A05_med	45.99	42.40	0.94	-2.12	73.01
A05_q05	76.66	73.33	0.94	-32.78	104.97
A05_q95	60.66	58.94	0.94	-16.78	91.55
DD	43.82	40.92	1.00	0.00	68.73
NEGW	45.74	41.96	0.94	-1.87	72.44

Common-mask ranking (full-coverage candidates):

Candidate	n_common	plain SMAPE	floor50 SMAPE
DD	34672.00	43.87	40.92
NEGW	34672.00	45.74	41.96
A05_med	34672.00	45.99	42.40
A05_huber	34672.00	57.02	56.43
A05_q95	34672.00	60.66	58.94

A05_q05	34672.00	76.66	73.33
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****Legal Oracle (calibration):**** plain sMAPE = ****16.40****, floor50 = 13.44, 9-16 bucket = 24.74, n_rows = 34672, invariant_pass = True.

5. Full Replay — New Candidate Tracks A–F

Candidate	plain sMAPE	floor50 sMAPE	cov.ratio	Δ vs DD (plain)	9-16 plain
A_QRA	46.84	43.88	0.94	-2.96	75.99
A_q05	76.66	73.33	0.94	-32.78	104.97
A_q50	41.90	38.95	0.94	1.97	65.75
A_q95	60.66	58.94	0.94	-16.78	91.55
B_midday	74.39	65.67	0.31	-5.56	74.39
B_midday_full	45.56	41.96	1.00	-1.75	73.97
C_seasonal_full	47.67	43.93	0.92	-3.63	75.41
C_shoulder	46.39	42.43	0.48	-3.17	75.12
C_summer	40.39	38.09	0.23	-2.87	71.42
C_winter	58.28	53.52	0.21	-5.48	80.33
DD	43.82	40.92	1.00	0.00	68.73
D_anchor	45.67	41.93	0.94	-1.79	72.12
E_fair	69.45	68.68	0.94	-25.58	115.28
E_huber	57.02	56.43	0.94	-13.15	95.91
F_regime	46.77	43.25	0.94	-2.90	73.72

****Common-mask ranking (full-coverage candidates):****

Candidate	n_common	plain sMAPE	floor50 sMAPE
A_q50	33893.00	41.96	38.98
DD	33893.00	44.04	41.06
D_anchor	33893.00	45.74	41.96
B_midday_full	33893.00	45.80	42.06
F_regime	33893.00	46.79	43.25
A_QRA	33893.00	46.98	43.98
C_seasonal_full	33893.00	47.67	43.94
E_huber	33893.00	57.39	56.80

A_q95	33893.00	61.04	59.29
E_fair	33893.00	69.65	68.89
A_q05	33893.00	76.92	73.62

Legal Oracle (new candidates): plain sMAPE = **13.34**, floor50 = 10.61, 9-16 bucket = 19.35, n_rows = 33893, invariant_pass = True.

6. Oracle Invariants

- Calibration oracle: eq_selected==candidate = True, loss ≤ each candidate = True, pass = True.
- New-candidate oracle: eq_selected==candidate = True, loss ≤ each candidate = True, pass = True.

7. Conclusion

All 8 V3.1 defects are corrected and guarded by contract tests. The corrected full 2022–2026 rolling-origin replay re-establishes the candidate metrics on a leak-free, business-day-consistent, metric-consistent basis. The legal Oracle remains an EX-POST upper bound (not deployable) and is used only to bound the best achievable per-row loss. The headline figures above **replace** the INVALIDATED_BY_V3.1_R1 numbers; they must not be compared to the old 64.84 / 41.39 / 145.71 values.

8. Reproduction

```
python tools/research/build_full_history_panel.py
python tools/research/full_history_replay.py # calibration
python tools/research/new_candidates_replay.py # tracks A-F
python -m pytest tests/research/ -v # 35 contract checks
python tools/research/generate_v31_report.py # this report
```